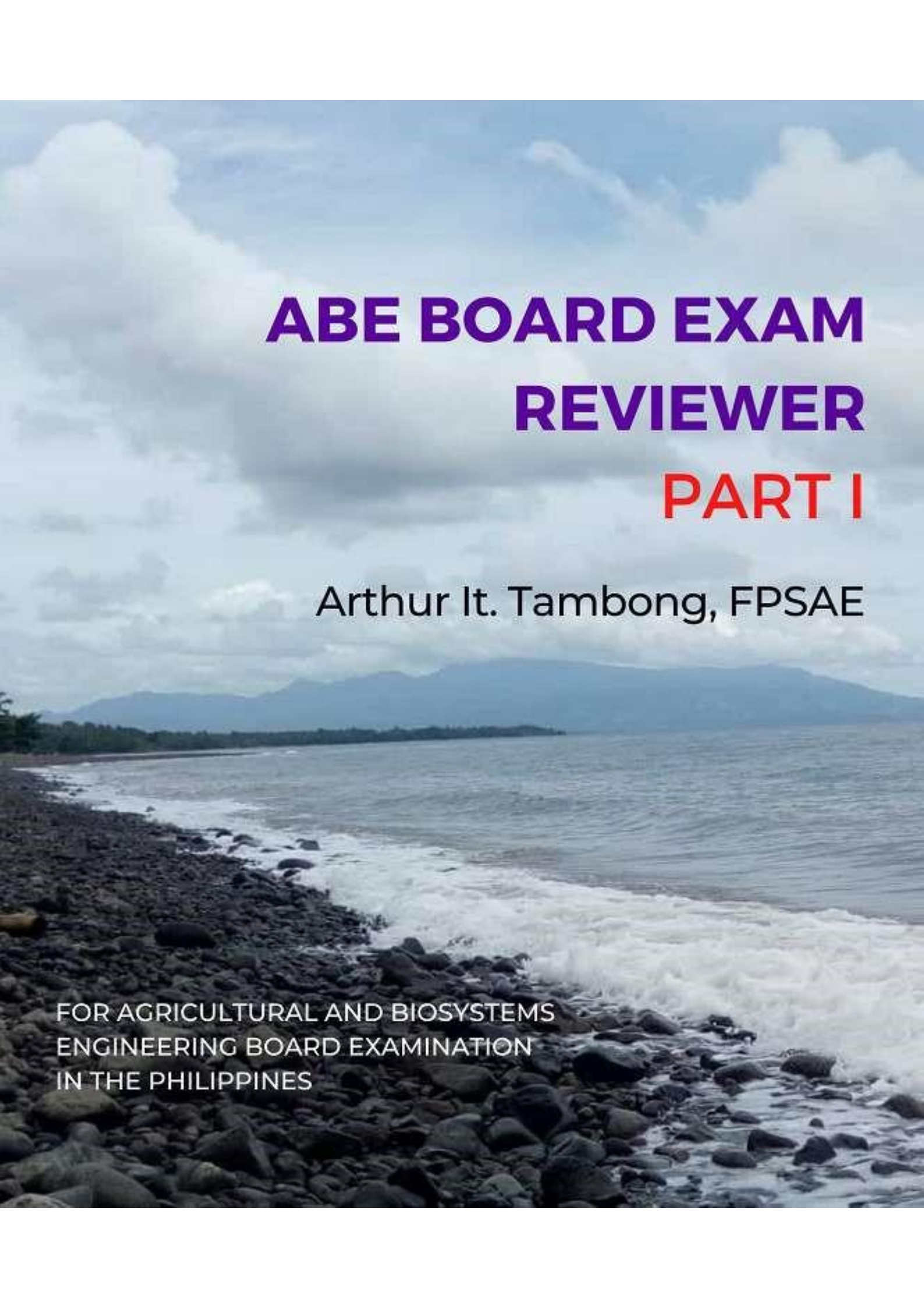




ABE BOARD EXAM REVIEWER

PART I

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FOR AGRICULTURAL AND BIOSYSTEMS
ENGINEERING BOARD EXAMINATION
IN THE PHILIPPINES

ABE BOARD EXAM REVIEWER

Part I

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Part I

Agricultural Mechanization, Power, Machinery and Equipment and Allied Subjects (33%)

TARGET COVERAGE OF THIS PART BASED ON PRC TABLE OF SPECIFICATIONS

- I. Agricultural Power and Tractor Operation (24%)

- II. Agricultural Machinery Management, Extension and Marketing (23%)
- III. Agricultural Machinery Design, Fabrication/Manufacturing and Testing (23%)
- IV. Agricultural Economics and Marketing (20%)
- V. AE Laws, Professional Ethics, Engineering Contracts, Specifications and Legal Documents (10%)

ESTIMATED COMPOSITION OF BOARD EXAMINATION QUESTIONS BY DEGREE OF DIFFICULTY

Easy – 40%
Moderate – 30%
Difficult – 30%

Review Questions

INSTRUCTIONS:

Select the best answer.
Give only one answer for each question.

I. Agricultural Power and Tractor Operation

A. GENERAL

EASY QUESTIONS:

1. The time rate of doing work or delivering energy, expressed as the amount of work done W or transferred energy divided by the time elapsed t or computed as W/t .

A. Force
B. Power
C. Work
D. Energy

Answer: B
2. What is the battery nominal voltage of a 4-wheel 90-hp tractor?

A. 12 Vac
B. 24 Vdc
C. 9 Vdc
D. 12 Vdc

Answer: D
3. The battery of a 24-volt combine harvester needs to be replaced. Two 12-volt batteries are to be used as its replacement. What should be the connection of the batteries?

A. Series
B. Parallel
C. Series-Parallel
D. Any connection

Answer: A
4. The 12-volt battery of a boom sprayer was drained. A 12-volt battery attached to a vehicle can be used for starting the engine. Jumper wires are available. What should be the connection of the two batteries to facilitate starting?

A. Series B. Parallel C. Series-Parallel D. Any connection
Answer: B

5. A tractor cannot start due to defective battery. It is being pushed by workers while lever is set to first gear. What should be foot-pressed and released abruptly to start engine?

A. Brake B. Clutch C. Clutch and Brake D. Accelerator
Answer: B

6. To gradually move a tractor from stop, what should be released while pressing the accelerator?

A. Brake B. Clutch C. Clutch and Brake D. Lever
Answer: B

7. To roll the tractor set at first gear in a flat land, what should be foot-pressed?

A. Brake B. Clutch C. Clutch and Brake D. Accelerator
Answer: D

8. To slow down the tractor running at first gear in a flat land, what should be foot-pressed?

A. Brake B. Clutch C. Clutch and Brake D. Accelerator
Answer: B (Clutch is better than clutch and Brake since purpose is only to slow down)

9. To gradually stop a fast running tractor, what should be foot-pressed?

A. Brake B. Clutch C. Clutch and Brake D. Accelerator
Answer: C

10. What is the safest gear setting for a descending tractor in a 10% slope?

A. Neutral B. Third gear C. Second gear D. First gear
Answer: D

11. In a routinary check before using the tractor, what should be checked first?

A. Fuel B. Water C. Brake D. Oil
Answer: C

12. To roll a tractor from stop and neutral position with engine off in a sloping land, what should be foot-pressed?

A. Brake B. Clutch C. Clutch and Brake D. Lever
Answer: B

13. The connection to the tractor of an erratic battery is to be stopped quickly. What battery terminal should be disconnected first?

A. Positive B. Negative C. Both terminals D. Any terminal
Answer: A

14. A new battery is to be installed to a tractor. What battery terminal should be connected first?

A. Positive B. Negative C. Both terminals D. Any terminal

Answer: B

15. The battery drains while tractor is not in use. What battery terminal should be disconnected when tractor is not in use?

A. Positive B. Negative C. Both terminals D. Any terminal

Answer: A

16. What kind of hitch is best for a non-hydraulic disk plow?

A. 1-point hitch B. 2-point hitch C. 3-point hitch D. 4-point hitch

Answer: C

17. What kind of hitch is best for a tractor trailer?

A. 1-point hitch B. 2-point hitch C. 3-point hitch D. 4-point hitch

Answer: A

18. A 24-volt battery of a pineapple harvesting conveyor needs to be charged on field using 12-volt solar modules. What is the minimum number of solar modules required?

A. One B. Two C. Three D. Four

Answer: B

Solution: Minimum no. of modules = No. of solar modules in series = $24v/12v = 2$ modules

19. A 12-volt battery of a boom sprayer needs to be charged on field using 6-volt solar modules with 2 modules in parallel to speed up charging. How many solar modules are required?

A. Two B. Four C. Six D. Eight

Answer: B

Solution:

No. of solar modules in series = $12v/6v = 2$ modules

Total no. of solar modules = Modules in series x Modules in parallel = $2 \times 2 = 4$ modules

MODERATE QUESTIONS:

20. What is the actual horsepower of a tractor which can do 149.2 kw-hrs of work in 2 hours?

A. 74.6 hp B. 57.5 hp C. 100 hp D. 105 hp

Answer: C

Solution:

$P = W/t \times \text{conversion factor}$

$= 149.2 \text{ kw-hrs}/2\text{hrs} \times 1 \text{ hp}/0.746 \text{ kw}$

$= 100 \text{ hp}$

21. What is the electrical energy consumption of a 30-kw vermin cast production shredder plugged in a 220-volt source if it requires 3 hours to shred 3 tonnes of farm waste?

A. 0.02 kwh/kg

B. 0.03 kwh/kg

C. 0.05 kwh/kg

D. 0.08 kwh/kg

Answer: B

Solution:

$$\begin{aligned} E_c &= \frac{P_c T_c}{W_i} \\ &= 30 \text{ kw } (3 \text{ h}) / 3,000 \text{ kg} \\ &= 0.03 \text{ kwh/kg} \end{aligned}$$

B. WIND POWER

EASY QUESTIONS:

1. In an aerogenerator exceeding 100 kw capacity, which part is responsible in converting rotational movement to electricity?

A. Rotor

B. Turbine

C. Alternator

D. Governor

Answer: C

2. What component of force is used in multi-bladed windmills?

A. Lift force

B. Drag force

C. Side force

D. Drop force

Answer: A

3. What component of force is used in savonius windmills?

A. Lift force

B. Drag force

C. Side force

D. Drop force

Answer: B

4. The type of pump is appropriate for multi-bladed windmills.

A. Hydraulic

B. Pneumatic

C. Reciprocating

D. Diaphragm

Answer: C

5. While the windmill is in operation, what efficiency can exceed 100%?

A. Volumetric

B. Transmission

C. Rotor

D. All of the above

Answer: A

6. What wind energy resource is located about 10-16 km from Earth surface with velocity over 150 kph?

A. Terrestrial wind

B. Extra-terrestrial wind

C. Jet stream

D. High wind

Answer: C

7. In multi-bladed windpumps, which part is responsible in converting rotational movement to reciprocating action?

A. Rotor

B. Connecting rod

C. Reciprocating pump

D. Crank arm

Answer: B

8. What tail type of multi-bladed windpump is more effective for windy areas?

A. Hinged side vane

B. Ecliptic

C. Manual folding

D. No tail

Answer: A

MODERATE QUESTIONS:

1. In wind farm development, which should be done first?
A. Rotor B. Energy demand survey C. Transmission D. Wind velocity measurement
Answer: D
2. In multi-bladed windpump manufacturing, which should be done first?
A. Rotor blade B. Rotor frame C. Tower D. Jig
Answer: D
3. To have bigger water displacement volume in windpumps, which should be increased?
A. Connecting rod length B. Rotor diameter C. Crank arm D. Well diameter *Answer: C*
4. In windmill design, what should be the average rotor blade twist angle in degrees to have a tip speed ratio of 1:1?
A. 30 B. 45 C. 60 D. 75
Answer: B
5. If typical overall efficiency of savonius windmills is 7-10%, what is the typical overall efficiency of aerogenerators?
A. 7-10% B. 15-17% C. 18-21% D. 90-95%
Answer: D

DIFFICULT QUESTIONS:

1. If a multi-bladed windpump has a tip-speed ratio of 1:1, what windspeed is needed to have its rotor tip move at 3.2 m/s?
A. 1 m/s B. 3 m/s C. 1 ft/s D. 3 ft/s
Answer: B

Solution:

$$\begin{aligned}\text{Windspeed} &= \text{tip speed, since tip-speed ratio is 1:1} \\ &= 3.2 \text{ m/s}\end{aligned}$$

C. SOLAR POWER

EASY QUESTIONS:

1. Which battery is designed for slow charging and slow discharging?

A. Maintenance-free battery B. Automotive battery C. Solar battery D. All of the above
Answer: C

2. Kind of battery useful in solar-powered energy systems.

A. Lead-acid battery B. Automotive battery C. Solar battery *Answer: D. All of the above*
D

3. Material used in solar cells.

A. Silicon B. Boron C. Equal silicon and boron D. Silicon with little boron
Answer: D

MODERATE QUESTIONS:

1. In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar pump without battery when load is in turned on?

A. 20-22 V B. 16-19 V C. 14-15 V D. 11-13 V
Answer: B

2. In a sunny 12:00 noon in the Philippines, what is the likely voltage reading of solar battery charger when no battery is being charged?

A. 20-22 V B. 16-19 V C. 14-15 V D. 11-13 V
Answer: A

3. At 8:00 P.M. in the Philippines, what is the likely voltage reading of a solar home system when all loads are turned on?

A. 20-22 V B. 16-19 V C. 14-15 V D. 11-13 V
Answer: D

DIFFICULT QUESTIONS:

1. In solar panel installation, which positive polarity wire should be connected last to the battery control unit?

A. Solar module B. Load C. Battery D. Monitor
Answer: C

2. If a coconut tree shades half of solar module, which condition is true?

A. Zero voltage B. Low voltage C. Zero amperage D. Low amperage
Answer: D

D. HYDRO POWER

EASY QUESTIONS:

1. A 260 kw hydro power plant is classified as

A. Microhydro B. Minihydro C. Small hydro D. Big hydro
Answer: B

2. Under Philippine classification, what is the maximum power rating for microhydro?

A. 25 kw B. 50 kw C. 75 kw D. 100 kw
Answer: D

II. Agricultural Machinery Management, Extension and Marketing

EASY QUESTIONS:

1. Refers to machines collectively.

A. Equipment B. Tools C. Machinery D. Machineries
Answer: C

2. The process of dealing with or controlling things.

A. Management B. Managing C. Supervising D. Administering
Answer: A

3. The action of promoting and selling products or services, including market research and advertising.

A. Certifying B. Marketing C. Endorsing D. Recommending
Answer: B

III. Agricultural Machinery Design, Fabrication/Manufacturing and Testing

EASY QUESTIONS:

1. Which one is a part of a vertical feed mixer?

- A. Screw B. Hopper C. Discharge chute D. All of the above
 Answer: D

2. A feed mixer tested to have mixing coefficient of variation of 10-15% is rated as

- A. Poor B. Fair C. Good D. Very good

Answer: C

3. As a standard, stripping efficiency of the mechanized stripping machine shall be at least

- A. 70% B. 80% C. 85% D. 90%

Answer: B

MODERATE QUESTIONS:

1. A 6-bladed windmill with 8 meters rotor diameter and 45 degrees twist angle is to be modified due to insufficient starting torque. What should be done?

- A. Increase blades B. Decrease blades C. Increase twist angle D. Decrease twist angle
 Answer: A
(since increasing blades increases torque)

2. A windmill with 6 meters rotor diameter, 5 cm pump diameter and 2 cm crank arm is to be modified due to low volumetric efficiency. What should be done?

- A. Enlarge rotor B. Reduce rotor C. Elongate crank arm D. Enlarge pump diameter

Answer: C *(since elongating crank arm increases pump stroke and volumetric efficiency)*

3. A 3-bladed 40-meter tall aerogenerator with 20 meters rotor radius designed for 6 m/s wind can't rotate. Average windspeed at rotor center is 5.5 m/s. Which part needs to be increased?

- A. Number of blades B. Rotor radius C. Crank arm D. Tower height

Answer: D *(windspeed is higher at higher elevation; increasing rotor radius will severely affect twist angle of entire blade & tip-speed ratio; aerogenerators have no crank arm)*

DIFFICULT QUESTIONS:

1. A 2-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 3 minutes. It produced 750 kg of crude oil in 3 hours. What is its crude oil production rate?

- A. 300 kg/h B. 250 kg/min C. 600 kg/h D. 600 kg/min

Answer: B

Solution:

$$P_{oil} = \text{Weight of crude oil collected} / \text{Total time} \\ = 750 \text{ kg} / 3 \text{ h} = 250 \text{ kg/h}$$

2. A castor bean oil expeller was loaded with 4.5 tonnes dried castor bean seeds. It produced 2,250 kg of crude oil in 4 hours. What is its castor bean cake production rate?

- A. 653 kg/h B. 563 kg/h C. 365 kg/h D. 635 kg/h

Answer: B

Solution:

$$\begin{aligned} P_{\text{cake}} &= \text{Weight of resulting cake produced} / \text{Total time} \\ &= (4,500 \text{ kg seed} - 2,250 \text{ kg oil}) / 4 \text{ h} \\ &= 562.5 \text{ kg/h} \end{aligned}$$

3. A 10-tonne capacity coconut oil expeller was loaded full with grated copra. It produced 5,000 kg of crude oil in 10 hours. What is the copra cake production rate of the machine at 95% cake collection efficiency?
- A. 745 kg/h B. 475 kg/h C. 547 kg/h D. 574 kg/h

Answer: B

Solution:

$$\begin{aligned} P_{\text{cake}} &= [(Wt. \text{ of copra input} - Wt. \text{ of crude oil collected}) / \text{Total time}] \times \text{Eff} \\ &= [(10,000 \text{ kg} - 5,000 \text{ kg}) / 10 \text{ h}] \times 0.95 \\ &= 475 \text{ kg/h} \end{aligned}$$

4. What is the weight of agricultural waste that is needed annually to optimally operate a 26-ampere 6-kw vermi cast production shredder having a capacity of 0.3 tonne of waste per hour? It optimally operates 16 hours daily during the 4-month peak sunny months and 8 h/day at 26 days/month for the rest of the year.
- A. 1,057 tonnes B. 1,075 tonnes C. 1,750 tonnes D. 175 tonnes

Answer: C

Solution:

$$\begin{aligned} Wt &= \text{Input Rate} \times (\text{Peak Operating Time} + \text{Lean Operating Time}) \\ &= [0.3 \text{ tonne/h}] [(16 \text{ h/d} \times 30 \text{ d/mo} \times 4 \text{ mo/yr}) + (8 \text{ h/d} \times 26 \text{ d/mo} \times 8 \text{ mos/yr})] \\ &= 1,075 \text{ tonnes/yr} \end{aligned}$$

IV. Agricultural Economics and Marketing

EASY QUESTIONS:

1. The benefits to society regardless of who in the society receives the benefits.

A. Financial benefit B. Economic benefit *Answer: B* C. Revenue D. Sales

2. Benefits less costs excluding depreciation.

A. Profit B. Net benefit C. Cash flow D. Economic benefits
Answer: C

3. Which one is a one-time expense?

A. Investment B. Single cost C. One-time cost D. Operating cost
Answer: A

4. Annually recurring cost.

A. Investment B. Operating cost C. Contingency D. Operating capital
Answer: B

5. Land is an example of

A. Investment B. Operating cost C. Contingency D. Operating capital
Answer: A

6. Manager's salary is an example of

A. Investment B. Operating cost C. Maintenance cost D. Operating capital
Answer: B

7. The cash component of investment.

A. Cash investment B. Operating cost *Answer: D* C. Contingency D. Operating capital

8. Contingency is normally what percentage of investment cost?

A. 5% B. 10% C. 15% D. 20%
Answer: B

9. The scrap value of a property?

A. Salvage value B. Economic value C. Book value D. End value
Answer: A

10. What is market price deducted by tax.

A. Discounted price B. Taxed price *Answer: C* C. Shadow price D. Economic price

11. The equivalent value of a property at any given year.

A. Salvage value B. Economic value C. Book value D. End value
Answer: C

12. Using simplest computation method, what is the annual depreciation of a tractor purchased at Php 2,500,000 if its life span is assumed at 8 years? Salvage value is 10%.

A. Php 218,250 B. Php 281,250 C. Php 821,250 D. Php 250,281

Answer: B

Solution: Depreciation = (Php2,500,000-Php250,000)/8 years = Php 281,250/year

13. What feasibility indicator is best for determining how fast the project recovers the investment?

A. NPV B. IRR C. BCR D. Payback period
Answer: D

14. What feasibility indicator is best for determining how much can you gain from a certain project?

- A. NPV B. IRR C. BCR D. Payback period

Answer: A

15. What feasibility indicator is normally compared to bank's interest rate?

- A. NPV B. IRR C. BCR D. Payback period

Answer: B

16. Which among the feasibility indicators is best for determining project performance?

- A. NPV B. IRR C. BCR D. Payback period

Answer: A

17. What analysis is done to determine the degree of project feasibility at various scenarios?

- A. Financial analysis B. Economic analysis C. Risk analysis D. Sensitivity analysis

Answer: D

18. What feasibility analysis uses shadow prices?

- A. Financial analysis B. Economic analysis C. Risk analysis D. Sensitivity analysis

Answer: B

19. What feasibility analysis uses market prices?

- A. Financial analysis B. Economic analysis C. Risk analysis Answer: A D. Sensitivity analysis

20. Which situation is better and attainable for a good economy?

- A. High discount rate B. Low discount rate C. Fast discount rate D. No discount rate

Answer: B

21. The normal combined discount and inflation rate of the Philippines.

- A. 2-4% B. 5-7% C. 8-12% D. 16-20%

Answer: C

22. The discounted cash flow at project start is always equal to ____.

- A. Zero B. Net benefit C. Net profit D. Net cash flow

Answer: D

23. The price at which there is no loss or gain.

A. Discounted price B. Optimum price C. Minimum price *Answer: D* D. Break-even price

MODERATE QUESTIONS:

1. The discount rate when the Net Present Value is zero.

A. Break-even rate B. IRR C. BCR D. ROI
Answer: B

2. What is the effect of increasing inflation rate to discounted cash flow at project start?

A. Increases B. Decreases C. Fluctuates D. None
Answer: D

3. What is the effect of increasing discount rate to discounted cash flow at year 10 of the project?

A. Increases B. Decreases C. Fluctuates D. None
Answer: B

4. Which one is true when discount rate increases?

A. NPV increases B. NPV decreases C. Risk decreases D. Capital reduces
Answer: B

5. At project start, the net cash flow of the cassava processing venture is -7 million pesos due to loan and revenue is not materialized yet. Discount rate is 10%. What is the value of the discounted cash flow in million pesos at project start?

A. 7 B. -7 C. -7.07 D. -7.7
Answer: B (At project start or time=0, the discounted cash flow is also equal to the net cash flow)

DIFFICULT QUESTIONS:

1. Using financial feasibility analysis, the Net Present Values (NPVs) of farm-to-market road, irrigation and flood control projects are 22.5, 7.5 and -3.1 million pesos, while the Internal Rate of Returns (IRRs) are 15%, 7% and 2.1%, respectively. Annual discount plus inflation rate is 10%. Which project shall be implemented?

A. Farm-to-market road B. Irrigation C. Flood control D. None
Answer: A (since only Farm-to-Market road project has IRR higher than discount plus inflation rate)

2. Using straight line depreciation method, what is the book value at the end of the third year of a tractor purchased at Php 2,500,000 if its life span is assumed at 10 years? Salvage value is 10%.

A. Php1,285,000 B. Php1,582,000 C. Php1,825,000 D. Php1,852,000

Answer: C

Solution:

Depreciation = (Php2,500,000-Php250,000)/10 years = Php 225,000/year

Book value = Purchase price – total depreciations = Php2,500,000-(Php225,000/yr x3yrs) = P1,825,000

3. Using economic feasibility analysis, the Net Present Values (NPVs) of small water impounding project (SWIP), shallow tubewell, small scale irrigation project (SSIP) are -0.5, -0.9 and 0 million pesos, respectively. The proposed projects are owned by the government. Which project is economically feasible?

- A. None B. SWIP C. Shallow tube well D. SSIP

Answer: D (In econ. analysis, NPV of 0 is feasible since it has economic benefits even if it is just break-even)

4. Microhydro, solar electrification and covered lagoon biodigester projects are evaluated for financial feasibility. Each requires PHP 20,000,000 investment. Only one project can be funded. The financial internal rates of return are 15%, 9% and 8%, respectively. Annual discount plus inflation rate is 10%. Which project shall be implemented?

- A. Covered lagoon biodigester B. Microhydro C. Solar electrification D. None *Answer: B (since its IRR is higher than discount plus inflation rate)*

5. A banana production and export project is being evaluated for financial feasibility. At 12% annual discount rate, it has the following feasibility indicators: net present value of 1.8 billion pesos and internal rate of return of 58%. If annual discount rate drops to 9%, will it be feasible?

- A. No B. Yes C. Maybe D. Insufficient data to determine

Answer: B (if it is feasible at 12% discount rate, the more it will be feasible at 9%)

V. AE Laws, Professional Ethics, Engineering Contracts, Specifications and Legal Documents

EASY QUESTIONS:

1. In what year was the Implementing Rules and Regulations (IRR) of the Philippine Agricultural and Biosystems Engineering Act of 2016 approved?

- A. 2016 B. 2018 C. 2020 D. 2022

Answer: D

2. What law is known as the Philippine Agricultural and Biosystems Engineering Act of 2016?

- A. RA 9513 B. RA 10601 C. RA 10915 D. RA 9736

Answer: C

3. Agricultural and Biosystems Engineering refers to the application of _____ and designs to the processes and systems involved in the sustainable production, post production and processing of safe food, feed, fiber, timber and other agricultural and biological materials and the efficient utilization, conservation, and management of natural and renewable resources in order to enhance human health in harmony with the environment.

- A. Agriculture B. Engineering C. Science D. Engineering science

Answer: D

4. In ABE law, agricultural and biosystems consist of crops, forestry and other plants, poultry, livestock, fisheries and _____ resources and other animals, wildlife and other living things.

- A. Agriculture B. Aquaculture C. Farm D. Land

Answer: B

5. Agricultural and Biosystems Engineer refers to a person who is registered and licensed to practice agricultural and biosystems engineering in the country and who holds a _____ and professional identification card from the Board of Agricultural and Biosystems Engineering and the PRC.

- A. Degree B. BSABE degree C. Experience D. Valid certificate of registration

Answer: D

6. Agricultural and Biosystems Automation and Instrumentation refers to the use and application of agricultural and biosystems _____.

- A. Circuits B. Instruments C. Sensors D. Integrated circuits

Answer: C

7. Agricultural and Biosystems Resource Conservation and Management refers to measures of conservation and proper management of agricultural and _____ resources.

- A. Biological B. Zoological C. Farm D. Water

Answer: A

8. In RA 10915, agricultural and _____ refers to local activity or series of activities to maintain or raise the quality or change the form or characteristics of agricultural, fishery, forestry and biological products/materials.

- A. Quality control B. Forestry C. Bio-processing D. Fishery

Answer: C

9. Agricultural and _____ System refers to utilization of information systems, database, and other information management tools for agricultural use, biological systems modeling to understand the mutual response between life and the environment.

- A. Computer B. Geographic Information C. Modeling D. Bio-information

Answer: D

10. Agricultural and Biosystems Power and Machinery refers to farm power and machinery for the production, harvesting, processing, storage, manufacture, preserving, transporting and _____ of agricultural and biological products/materials.

- A. Distribution B. Quality control C. Marketing D. Sales

Answer: A

11. Agricultural and Biosystems Buildings and Structures refer to buildings and structures for the production, harvesting, processing, storage, _____, preserving, transporting and distribution of agricultural and biological products/materials.

- A. Treatment B. Manufacture C. Quality control D. Sales

Answer: B

12. In Section 36 or ABE law, the existing Philippine Agricultural Engineering Standards (PAES) shall be transformed into a Philippine Agricultural and Biosystems Engineering Standards and shall serve as Code of _____ Standards.

- A. AE B. ABE C. Technical D. Engineering

Answer: C

13. The ____ earned by the professional shall be required in the renewal of professional license.

- A. Degree B. BSAE degree C. BSABE degree D. CPD credit units

Answer: D

14. Any person who violates any of the provisions of RA 10915, upon conviction, shall be penalized by a fine of not less than P100,000 but not more than ____, or imprisonment of not less than 6 months but not more than 5 years, or both fine and imprisonment, at the court discretion.

- A. P200,000 B. P300,000 C. P400,000 D. P500,000

Answer: D

15. In Section 17 of RA 10915, which one is not a scope of Land and Water Resources Engineering?

- A. Irrigation B. Soil and water C. Aquaculture D. Meteorology

Answer: D

16. Which republic act is known as the Philippine Agri-Fishery Mechanization Law?

- A. RA 9513 B. RA 10915 C. RA 10601 D. RA 9736

Answer: C

17. Under RA 10915, which one is not under the scope of Land and Water Resources Engineering?

- A. Irrigation B. Soil and water C. Aquaculture D. Meteorology

Answer: D

18. Under Washington Accord, an Agricultural and Biosystems Engineer who passed licensure examination in the Philippines can practice his/her profession in another Washington Accord member country.

- A. Yes B. No C. Indeterminate D. Insufficient data to determine

Answer: A

19. In Washington Accord, the contract of ABE services signed in the Philippines is valid in another Washington Accord member country.

- A. Yes B. No C. Indeterminate D. Insufficient data to determine

Answer: B

20. As provided in Article V Section 20 of AFMech Law, all agricultural and fisheries machinery manufacturers, fabricators, assemblers, and importers must register with the _____.

- A. BAFE B. AMTEC C. DTI D. LTO

Answer: A

21. As provided in Article VI Section 27 of AFMech Law, the _____ is designated as the premier and reference testing center in the country for agricultural and fisheries machinery. _____.

- A. BAFE B. AMTEC C. IPO Philippines D. BAFS

Answer: B

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